Kirby Johnson

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EDUCATION

JOHNS HOPKINS UNIVERSITY — **Baltimore, MD**
M.S.E., Biomedical Engineering, Center for Bioengineering Innovation and Design (CBID) | GPA: 4.0/4.0 — Expected May 2027

UNIVERSITY OF CALIFORNIA, LOS ANGELES — **Los Angeles, CA**
B.S., Bioengineering | GPA: 3.7/4.0 | Dean's Honor Student — June 2026

PROFESSIONAL EXPERIENCE

JOHNS HOPKINS CENTER FOR BIOENGINEERING INNOVATION & DESIGN — **Baltimore, MD**
Microsurgery Biodesign Team, Industry-Sponsored Project — **June 2026 - Present**

- Conducted 180+ hours of clinical needs assessment in reconstructive microsurgery and peripheral nerve ORs to identify unmet device opportunities
- Validated 349 clinical needs through over 50 interviews with surgeons, value analysis committees, patients, and other stakeholders
- Evaluated clinical, technical, and commercial feasibility to select the highest-impact need for development
- Translate the selected need into a device through concept generation, rapid prototyping, and patent filing

Improving Joint Pain Care in Rural India, Industry-Sponsored Project — **June 2026 - Present**

- Identified barriers to musculoskeletal care through field research across 15 rural and urban healthcare sites in India
- Validated 367 findings with patients, care providers, and an open-source NGO partner
- Prioritize needs by clinical impact, technical feasibility, and market potential ahead of solution development
- Develop a solution to expand rehabilitation access for exercise-responsive MSK pain and test in follow-up field study

APPLIED MEDICAL — **Rancho Santa Margarita, CA**
Engineering Intern, Surgical Simulation — **June 2025 - September 2025**

- Designed and implemented a production test fixture improving camera alignment for AI-enabled surgical testing
- Optimized gallbladder simulator molding process, doubling part output per cycle and reducing manufacturing lead time
- Designed and executed Design of Experiments (DoE) to optimize ultrasonic welding equipment parameters
- Executed IQ/OQ validation and authored engineering documentation and SolidWorks manufacturing drawings

UCLA HEALTH: ADULT CONGENITAL HEART DISEASE — **Los Angeles, CA**
Clinical Research Coordinator — **October 2023 - Present**

- Managed 7 concurrent clinical studies of structural heart devices, managing patient enrollment, physician communication, and sponsor reporting
- Interface and coordinate with medical device sponsors to support patient-facing studies and report device performance observations
- Secured IRB and FDA approvals for 20 Compassionate Use stent procedures for ASD patients, independently preparing submissions
- Led major data collection effort for native RVOT procedures device FDA submission, with UCLA as primary site

UCLA ENGINEERING AMBASSADORS — **Los Angeles, CA**
Ambassador — **May 2025 - June 2026**

- Conducted campus tours and Q&A sessions for prospective Henry Samueli School of Engineering students and their families
- Represented the School of Engineering at admissions and outreach events

ENGINEERING PROJECTS

GAIT ANALYSIS WEARABLE - SENIOR CAPSTONE PROJECT — **Los Angeles, CA**
Team Engineer — **September 2025 - June 2026**

- Developed a wearable system integrating multiple IMUs and ML-enabled feedback to characterize Parkinson's gait abnormalities
- Led mechanical integration: optimized sensor placement and prototyped 3D-printed and soft-good housings
- Finalist, Rice 360° Undergraduate Global Health Technologies Design Competition

BIOMEDICAL ENGINEERING SOCIETY — **Los Angeles, CA**
Mechanical Subteam - Design Team — **October 2025 - June 2026**

- Designed a PEG-tube securement device that reduces dislodgement risk and internal tissue trauma during high traction events
- Tested prototypes in the UCLA Health Simulation Center, evaluating retention force and patient-comfort requirements in realistic clinical scenarios

SKILLS

CAD & Prototyping: Creo Parametric, SolidWorks, Fusion 360, AutoCAD, CNC machining, 3D printing, molding, Design for Manufacturing (DfM)
Hardware & Programming: IMU integration, PCB design, MATLAB, Arduino/C++, Python
Testing & Validation: Design of Experiments (DoE), IQ/OQ protocols, Engineering Evaluations, Work Instructions (WIs), Test Protocols, IRB submissions

LEADERSHIP & ENGAGEMENT

ENGINEERING TOMORROW - Presenting Engineer — **September 2026 - Present**

- Conduct hands-on engineering labs for high school students, guiding design challenges and sharing career pathways

SOCIETY OF WOMEN ENGINEERS AT JHU - Active Member — **September 2026 - Present**
JOHNS HOPKINS UNIVERSITY CLUB VOLLEYBALL - Team Member — **September 2026 - Present**

PUBLICATIONS

- "Long-Term Durability of Melody Valves Implanted in Small (< 30kg) Patients," Journal of the American College of Cardiology, May 2025